Date: _____

Question 1.

Input	Output
• rows	• seat Number & Symbols
• seats	• Initial & final seating arrangement
• consecutive seats	• Allocated rows & seats
	• Initial & final available seats.

Pseudocode:

```
PRINT ("Number of Seating Rows") INPUT Rows.
PRINT ("Number of Seats per Row") INPut Seats.

DECHARE seat_Arrangement [rows] [seats] : ARRAY.

PRINT ("Enter Seating Arrangement")
For { i = 0, i < rows
    For  j=0 , j < seats
        INPUT seat_Arrangement [i] [j]
```

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```
PRINT ("Seats: ")
FOR i = 0, i < seats)
    PRINT (i)

FOR (i = 0, i < rows)
    PRINT ("Row", i)

    FOR (j = 0, j < seats)
        IF seat_Arrangement[i][j] = 0
            PRINT ("_")
        ELSE PRINT ("X")
total = 0.
FOR (i = 0, i < rows)
    PRINT ("Rows", i), count = 0.
    FOR (j = 0, j < seats)
        IF (seat-Arrangement[i][j] = 0)
            count++, total++
    PRINT (Count, "are available")
PRINT ("Total Number of Seats", Total)

print ("Enter No. of Customers") INPUT K

IF (K > seats OR K <= 0) PRINT ("INVALID Size")
    Return -1
ELSE
```


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found = 0, startSeat = 0, row = 0, middle = row/2.

FOR (d = 0, d < rows AND found == 0)
// Finding correct ~~rows~~ seats from center
Used Binary Search Logic.
left = middle - d, right = middle + d.

IF (left >= 0)
consecutive = 0.
FOR (j = 0, j < seats)
IF (seat - Arrangement[left][j] == 0).
consecutive++
ELSE consecutive = 0.

IF (consecutive == 1) startSeat = j.
IF (consecutive == k)
row = left, found = 1, break

~~IF (right < rows && right != left && found == 0).~~

IF (right < rows AND found == 0).
consecutive = 0.

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```
FOR (j = 0, j < seats)        right
    IF (seat_Arrangement[i][j] == 0)
        consecutive ++
    ELSE consecutive = 0

    IF (consecutive == 1) startSeat = j
    IF (consecutive == k)
        row = right, found = 1, break

    IF (found == 1)
        PRINT ("Seats Booked")
        PRINT ("Allocated Row", row)
        PRINT ("Allocated Seats:")

        FOR (p = startSeat, p < startSeat + k)
            PRINT (p)
            seat_Arrangement[row][p] = 1

    ELSE PRINT ("No seats Available")

PRINT ("Seat")
FOR (i = 0, i < seats)
    PRINT (i)
```


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```
FOR (i=0, i<rows)
    PRINT ("Row:", +i).

FOR( int j=0, j<seats)
    IF (seat_Arrangement[i][j] = =0]
        print("c ")
    ELSE PRINT ("X")

PRINT ("Total Number of Seats', total_k)
```

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